13.10 Equivalent Linear Expressions

Lesson Introduction

Benchmark	MA.7.AR.1.2 Determine whether two linear expressions are equivalent.	Learning Target	• I can verify using substitution that linear expressions are equivalent.

	Building On	Working Toward
Benchmark Coherence	MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.	MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

Essential Questions	• What is the difference between equivalence and equality? • How can you determine whether two algebraic expressions are equivalent?
Lesson Narrative	Students extend their understanding of equivalent expressions to include substituting rational numbers into variable expressions and evaluating them to determine equivalence. The concept of equal versus equivalent is explored, and students debate what values are ideal to use when proving equivalence. The focus of this lesson should be on substitution, not simplifying expressions to prove equivalence.

Component Summary	13.10.1 Warm-Up (~5 min.)	Reflect on the origins of the word “algebra” *(SE p. 314)*	13.10.4 Guided Instruction (10 min.)	Determine whether algebraic expressions are equivalent using substitution *(SE p. 316)*
	13.10.2 Refresher (5 min.)	Evaluate expressions for given values of the variable using substitution *(SE p. 315)*	13.10.5 Guided Practice (10 min.)	Use substitution to verify if expressions are equivalent *(SE p. 318)*
	13.10.3 Collaboration (10 min.)	Evaluate expressions for multiple values to determine if the expressions are equivalent *(SE p. 315)*	13.10.6 Your Turn! (5 min.)	Determine which expressions are equivalent by substituting values for the variable *(SE p. 319)*
	13.10.7 Wrap-Up (~5 min.)	Self-assessments of understanding of lesson’s learning targets *(SE p. 320)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: • Equivalent • Equal <u>Additional Terms</u>: N/A	• (Optional) Colored pencils • (Optional) Chart paper • (Optional) Markers	• Consider providing an anchor chart, word wall, or other visual representation of the key terms in this lesson for visual learners.

Teacher Tips

Common Misconceptions	Things to Consider
• Students may add substituted values with their coefficients rather than multiplying. For example, in question 1 in 13.10.2 Refresher, students may add $-3(5)$ to get 2 rather than multiplying to get -15. • Students may describe equivalent expressions as “equal” to each other.	• Lesson 12 will revisit the use of properties as a method of determining equivalence, while this lesson targets opportunities for substitution, with the 13.10.7 Wrap-Up giving students choice in which method they choose. • Determine if your students need the support of all or part of the 13.10.2 Refresher • If time is an issue, then consider assigning 13.10.6 Your Turn! for homework.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Stronger and Clearer Students extend their understanding of equivalence from numeric to algebraic expressions, particularly making this connection during 13.10.3 Collaboration. Encourage students to engage in a Stronger and Clearer routine for articulating their responses in question 3, taking multiple attempts and getting feedback from their partner to make each attempt better, stronger, and clearer.	**MA.K12.MTR.1.1: Participate in Effortful Learning** Notice the elements of collaboration embedded throughout this lesson that provide opportunities for students to collectively contribute toward learning targets. Monitor student responses as they discuss their thinking in 13.10.3 Collaboration, 13.10.4 Guided Instruction, and 13.10.5 Guided Practice, and use talk moves to facilitate student discussion as they refine their responses and learn from each other.

Differentiate Instruction

Collaboration and discussion are key in this lesson. Utilize sentence frames to enhance student discussion to target key learning opportunities as students collaborate throughout the lesson. Sentence frames provide low-floor, high-ceiling entry points for learners to either enter the conversation or extend their thinking to incorporate mathematical concepts.

Expression___

Is equivalent to ________. I used a method of ______ to show that _____

substitution | evaluating | properties | equivalent

Lesson Notes:

13.10.1 Warm-Up: Origins of Algebra

Component Overview: *Students reflect on the origin of the word "algebra" and how it relates to algebraic expressions and equations.*

13.10.1 Warm-Up: Origins of Algebra

The word **algebra** comes from the Arabic word الجبر, or al-jabr, which means, "the restoring of broken parts." The word algebra comes from the title of a book written by a Persian mathematician named Muhammad ibn Musa al-Khwarizmi in the 9th century.

1. Based on what you know about algebraic expressions and equations, explain why you think this meaning of the Arabic word makes sense to describe algebra.

   Answers will vary.

   "Restoring of broken parts" makes sense for algebra because algebra includes solving equations and combining parts (terms) that are alike.

Students will provide a variety of creative answers for this prompt. Consider what connections could be made to content in this unit or in previous units.

13.10.2 Refresher: Evaluating Algebraic Expressions

Component Overview: *Students work to evaluate expressions for given values of the variable. Quickly review the values of each expression before moving on to ensure students have checked their work and are given time to ask clarifying questions.*

13.10.2 Refresher: Evaluating Algebraic Expressions

Evaluate each expression for the given value of x.

1. $-3x - 8$ for $x = 5$

   $= -3(5) - 8$
   $= -15 - 8$
   $= -23$

2. $4 - 2(x^2 - 7)$ for $x = -1$

   $= 4 - 2(1 - 7)$
   $= 4 - 2(-6)$
   $= 4 + 12$
   $= 16$

3. $(4.5 + x) \cdot 3$ for $x = -\frac{5}{2}$

   $= (4.5 + (-\frac{5}{2})) \cdot 3$
   $= (4.5 - 2.5) \cdot 3$
   $= 2 \cdot 3$
   $= 6$

Students continue practicing throughout the lesson. Avoid reteaching operations and substitution. Instead, make note of which students could use additional support and pull a small group during 13.10.3 Collaboration or 13.10.5 Guided Practice based on your observations.

13.10.3 Collaboration: Solving Equivalent Expressions Using Substitution

Component Overview: *Students work with a partner to evaluate expressions for given values and determine whether they are equivalent, reasoning that the expressions are equivalent because they have the same value when evaluated for multiple values. Consider engaging the class in a Notice and Wonder routine before completing question 4 and revisiting their wonderings after they have simplified both expressions.*

13.10.3 Collaboration: Solving Equivalent Expressions Using Substitution

1. With your partner, complete the table by choosing who will be Partner A or Partner B to determine if the expressions are equivalent.

Partner and Value	$8x + 4x - 2$	$2(6x - 1)$	Equivalent?
A $x = 2$	$= 8(2) + 4(2) - 2$ $= 16 + 8 - 2$ $= 22$	$= 2(12 - 1)$ $= 2(11)$ $= 22$	o yes o no
B $x = 5$	$= 8(5) + 4(5) - 2$ $= 40 + 20 - 2$ $= 58$	$= 2(30 - 1)$ $= 2(29)$ $= 58$	o yes o no
A $x = -5$	$= 8(-5) + 4(-5) - 2$ $= -40 + -20 + -2$ $= -62$	$= 2(-30 + -1)$ $= -62$	o yes o no
B $x = (-2)$	$= 8(-2) + 4(-2) - 2$ $= -16 + -8 + -2$ $= -26$	$= 2(-12 + -1)$ $= -26$	o yes o no

2. Complete the statement to describe the results in the last column of the table.

   Depending on the value being substituted into the expressions, the evaluated expressions had the same value as each other.
   - o always
   - o sometimes
   - o never

3. What do the results tell you about the expressions $8x + 4x - 2$ and $2(6x - 1)$?
   They are equivalent expressions.

4. Rewrite $8x + 4x - 2$ by combining the like terms.
   $12x - 2$

5. Use the distributive property to rewrite $2(6x - 1)$.
   $12x - 2$

6. Discuss with your partner how the two expressions verify the results from your table.

Consider having students who finish early make a prediction about future expressions.
- *How do the properties of operations relate to these expressions and their equivalence?*
- *How could properties of operations be used to determine equivalence without evaluating using substitution?*

This rationale and method of thinking builds proficiency in fluency (MA.K12.MTR.3.1).

Stronger and Clearer
Question 3 sample response: "Just like numeric expressions can look different but be the same, linear expressions can also look different but actually have the same value."

13.10.4 Guided Instruction: Determining Equivalent Expressions Using Substitution

Component Overview: *Students review the definitions and difference between equivalence and equality. Then students will determine whether expressions are equivalent by substituting different values for x. Students will be guided through why the expressions are not equivalent and consider why using zero as a variable value can be both beneficial and problematic.*

13.10.4 Guided Instruction: Determining Equivalent Expressions Using Substitution

The two expressions $8x + 4x - 2$ and $2(6x - 1)$ are equivalent, but not equal. The expressions $12x - 2$ and $12x - 2$ are equal.

1. What is the difference between **equivalence** and **equality**?

> Algebraic expressions are **equivalent** if they have different terms but are equal when evaluated using the same value for the variables. Algebraic expressions are **equal** if they have the same terms and operations.

This is a discussion opportunity. Students do not need to write an explanation here. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

Use talk moves to prompt students to make sense of the difference between equivalent and equal. Consider a Think-Pair-Share or similar routine and prompt students to include an example that illustrates their working definitions. Sample exemplar response: "Equivalent expressions have the same value. Equal expressions are exactly the same in appearance."

2. Determine whether $5 - 3(2x + 1)$ and $4x + 2$ are equivalent using substitution of numerical values.

   a. Complete the table.

x	$5 - 3(2x + 1)$	$4x + 2$
What is the value when $x = 0$?	$= 5 - 3(0 + 1)$ $= 5 + -3(1)$ $= 5 + -3$ $= 2$	$= 0 + 2$ $= 2$
What is the value when $x = \frac{1}{2}$?	$= 5 + -3(1 + 1)$ $= 5 + -3(2)$ $= 5 + -6$ $= -1$	$= 2 + 2$ $= 4$
Choose another value for x to substitute. *Answers will vary. For example, if we let $x = -2$.*	$= 5 + -3(-4 + 1)$ $= 5 + -3(-3)$ $= 5 + 9$ $= 14$	$= -8 + 6$ $= -2$

13.10.4 Guided Instruction: Determining Equivalent Expressions Using Substitution (continued)

b. Explain if the expressions are equivalent.
No, the expressions are not equivalent. They do not have the same value for all values substituted in for the variable.

c. Explain why the expressions were equal when "0" was substituted for x.
They were equal when 0 was substituted in for the variable, because the other terms of the expression are equivalent.

d. Discuss with your partner the advantages and disadvantages of using "0" to substitute for a variable to check for equivalence.
The advantages are that the calculations are easy. The disadvantages are that it may appear that the expressions are equivalent when they aren't.

3. Determine whether $3 + x(4 - 2)$ and $x(3 - 7) + 5$ are equivalent using substitution of numerical values.

a. Complete the table.

x	$3 + x(4 - 2)$	$x(3 - 7) + 5$
What is the value when $x = 1$?	$= 3 + (2)$ $= 5$	$= 1(-4) + 5$ $= 1$
What is the value when $x = 0$?	$= 3 + 0$ $= 3$	$= 0 + 5$ $= 5$
Choose another value for x to substitute. *Answers will vary. One example is $x = -2$.*	$= 3 + -2(4 - 2)$ $= 3 + -2(2)$ $= 3 + -4$ $= -1$	$= -2(3 - 7) + 5$ $= -2(-4) + 5$ $= 8 + 5$ $= 13$

b. How many values are best to use when substituting values in for the variable to check for equivalency?
It depends, but I like to substitute at least one negative and one positive number. If you only test zero, your calculations may be easy. However, the expressions may appear equivalent, when they are not.

Notice the multiple entry points for learning in this component. Utilize all entry points for students to engage visually with the table or verbally by explaining and discussing.

Consider asking students the following:

- *What are the advantages and disadvantages of substituting zero in for the variable?*
- *How many values do you need to substitute in for the variable to determine if the expressions are equivalent?*
- *Should you always use zero to determine equivalence? Why or why not?*

13.10.5 Guided Practice: Determining Equivalent Expressions Using Substitution

Component Overview: *Students work independently or in pairs to determine equivalence.*

13.10.5 Guided Practice: Determining Equivalent Expressions Using Substitution

1. Consider the three expressions below. Determine whether the expressions are equivalent by substituting the given value of x.

   a. Complete the table.

x	$4x-\frac{1}{2}(6+8x)$	$8x-3$	$-(4x+3)+4x$
$x=3$	$=12+-1/2(6+24)$ $=12+-1/2(30)$ $=12+-15$ $=-3$	$=24-3$ $=21$	$=-1(12+3)$ $+12$ $=-15+12$ $=-3$
$x=-2.75$	$=4(-2.75)+-1/2(6+8(-2.75))$ $=-11+-1/2(6+-22)$ $=-11+-1/2(-16)$ $-11+8$ -3	$=-22$ -3 $=-25$	$=-1(-11+3)$ $+-11$ $=-1(-8)+-11$ $=8+-11$ $=-3$
Answers will vary. For example, $x=1$.	$=4+-1/2(6+8)$ $=4+-1/2(14)$ $=4+-7$ $=-3$	$=8-3$ $=5$	$=-1(4+3)+4$ $=-7+4$ $=-3$

   b. What conclusion can you draw about the expressions from the information in the table?
   The expressions $4x-\frac{1}{2}(6+8x)$ *and* $-(4x+3)+4x$ *are equivalent. The expression* $8x-3$ *is not equivalent to either expression.*

Consider a small group or student choice based on the needs of your students.

- *Small Group: Pull a small group to model or think-aloud operations and properties with students identified from 13.10.2 Refresher.*
- *Choice: Allow students to choose which row to complete then find a partner around the room who completed the same row to compare responses. Additionally, students could choose their own value for x and substitute to prove equivalence.*

Ask advanced learners to rewrite the expression that is not equivalent to the other two expressions so that it is equivalent. Challenge them to include decimals and fractions and at least four different operations in their expression.

2. Determine whether the expressions $\frac{2}{5}(15x - 30y) + 10x$ and $4(4x - 2y)$ are equivalent.

x **and** y	$\frac{2}{5}(15x - 30y) + 10x$	$4(4x - 2y)$
$x = -2$ $y = 0$	$= \frac{2}{5}(-30) + -20$ $= -12 + -20$ $= -32$	$= 4(-8)$ $= -32$
Answers will vary. For example, $x = 1$ $y = 1$	$= \frac{2}{5}(15 - 30) + 10$ $= \frac{2}{5}(-15) + 10$ $= -6 + 10$ $= 4$	$= 4(4 - 2)$ $= 4(2)$ $= 8$

No, they are not equivalent.

Review the table and student responses to question 1b to ensure students are mastering the content. Correct any errors as necessary before moving on to question 2. This is an opportunity to encourage student precision in multiplication. Encourage students to check their work.

13.10.6 Your Turn!

Component Overview: *Students work independently to determine which expressions are equivalent by substituting values for the variable.*

13.10.6 Your Turn!

1. Determine which expressions are equivalent by substituting the same values for the variable.

A $(5x-3)-\frac{3}{4}(8+2x)$ $x=2$ $=(10-3)-\frac{3}{4}(8+4)$ $=7-\frac{3}{4}(12)$ $=7-9$ $=-2$	B $-5\frac{1}{4}(4x-1)$ $x=2$ $-\frac{21}{4}(8-1)$ $-\frac{21}{4}(7)$ $=-\frac{147}{4}$ $=-36\frac{3}{4}$
C $9-6\left(\frac{-2}{3}x+4\right)$ $x=2$ $=9+-6(-\frac{4}{3}+4)$ $=9+8+-24$ $=-7$	D $\frac{3}{2}x-9+2x$ $x=2$ $=3-9+4$ $=-6+4$ $=-2$
E $\left(\frac{30}{2}\right)(-1)+4x$ $x=2$ $=-15+8$ $=-7$	F $6-12\frac{1}{2}x-\frac{3}{4}-8.5x$ $x=2$ $=6+(-25)+-\frac{3}{4}+(-17)$ $=-36\frac{3}{4}$

Consider providing students with which expressions are equivalent as scaffolding. This acts as a self-check for students who need to build confidence and fluency in their calculations.

2. Complete each statement using the expressions from the table.

A is equivalent to D

C is equivalent to E

B is equivalent to F

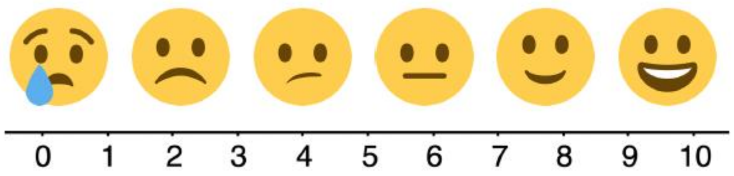
13.10.7 Wrap-Up: Reflection

Component Overview: *Students complete a self-assessment of their understanding of the lesson's learning targets using an emoji scale. Then students provide a written explanation to support their choice of emoji.*

13.10.7 Wrap-Up: Reflection

1. Rate yourself using the scale below on today's learning target.
   Answers will vary.

   I can verify using substitution that linear expressions are equivalent.



2. Write an explanation of why you rated yourself where you did.
   Answers will vary.

Consider allowing students to mark-up text and label each activity with an emoji. Students could also draw their own emoji for each lesson component.